Freight Rate Prediction

Data quality, time-based validation, model selection, and final outputs

Assessment snapshot

LABELED DATA

48,000 loads | Jan 1-Oct 31, 2025

FUTURE DATA

12,000 loads | Nov 1-Dec 31, 2025

SELECTED MODEL

LightGBM, without market signals

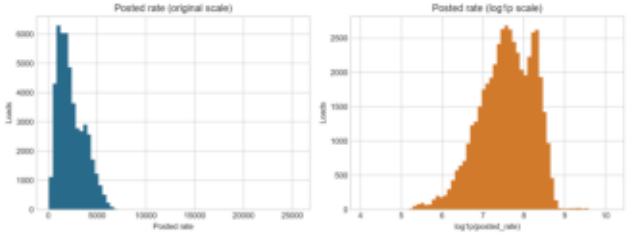
SUBMISSION STATE

Local format checks passed; Spotter metrics pending

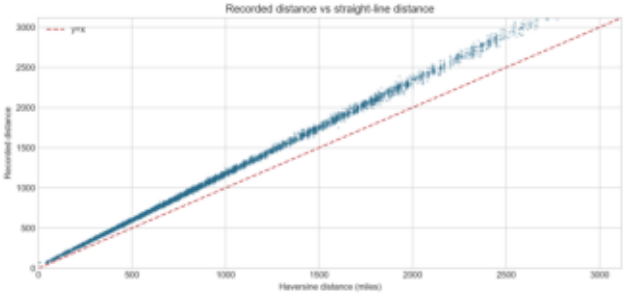
EDA findings and treatment

Check	Observed result	Modeling response
Weight quality	Train: 300 missing, 292 negative; validation: 165 missing	Use absolute values; median-impute within each training fo
Market index	374 train and 249 validation values missing; SMD is -1.20	Fit imputation on training rows; do not rely on a future sign
Target shape	Posted rate is right-skewed (skew 1.902); 260 IQR outlier	Train on log1p rate with a 99th-percentile cap learned per f
Geography	Coordinate-derived distance correlates strongly (0.9995)	Keep supplied road miles; retain haversine separately. City
Duplicates / route	No exact duplicate rows, duplicate IDs, or pickup=deliver	No removal required for these checks.

The market index and quote signal have weak marginal correlation with rate (Pearson 0.034 and -0.040). The future market-index distribution also shifts, so the signals are not assumed to be stable.



Posted-rate distribution



Supplied vs coordinate distance

Validation and model selection

Chronological evaluation; all reported values are from saved local outputs

Evaluation design

Holdout: train on Jan-Aug 2025 (38,477 rows), evaluate Sep-Oct (9,523 rows). Expanding-window monthly CV evaluates May, June, July, and August after training only on earlier dates. Pipelines fit imputers and encoders on each training partition. Random splitting would mix future and past conditions and overstate performance.

September-October holdout

Model	MAE	RMSE	MAPE	RMSLE
Median rate/mile	$256.95	$684.25	11.63%	0.1981
Ridge (log target)	$161.85	$651.88	6.68%	0.1656
LightGBM (no signals)	$126.99	$639.94	5.43%	0.1621
LightGBM (estimated signals)	$141.66	$644.01	6.11%	0.1643

The no-signals LightGBM has the best holdout score on all four reported metrics. The estimated-signals variant uses a linear extrapolation of recent weekly signal means, clipped to historical 1st-99th percentiles; it did not help here.

Four-fold mean (May-August)

Model	MAE	RMSE	MAPE	RMSLE
Median rate/mile	$245.93	$672.47	11.47%	0.1960
Ridge (log target)	$133.76	$631.67	5.57%	0.1551
LightGBM (no signals)	$134.05	$631.99	5.76%	0.1562
LightGBM (estimated signals)	$142.91	$633.83	6.17%	0.1577

Interpretation: Ridge and no-signals LightGBM are close on the expanding folds; LightGBM wins the untouched Sep-Oct holdout. Top LightGBM features include log distance, historical lane encoding, weight, haversine miles, and time trend. Error breakdowns and feature importance are saved under outputs/stage3/.

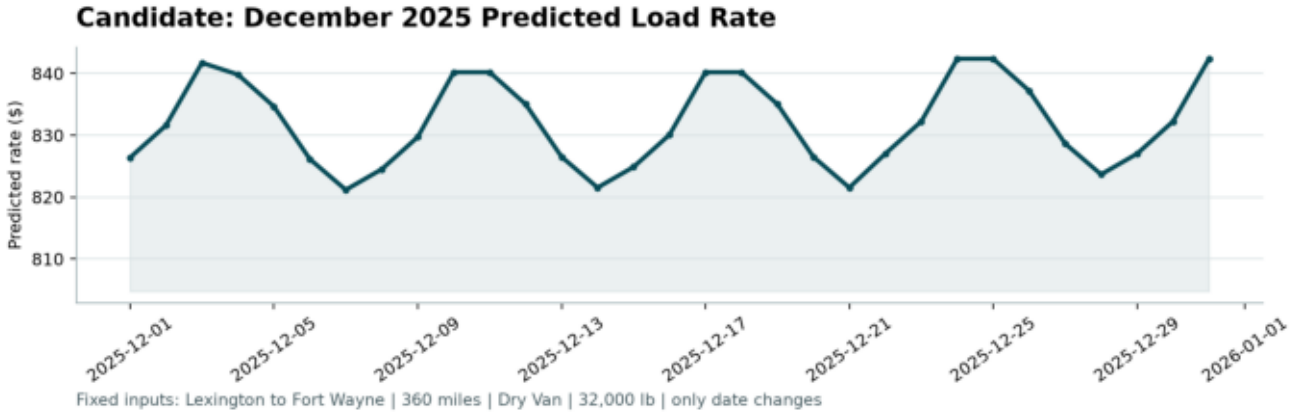
Final outputs and caveats

Candidate December curve for the fixed Lexington-to-Fort Wayne lane

Chosen approach

The chosen no-signals LightGBM is retrained on all 48,000 labeled loads. It uses distance, weight, equipment, route, time, haversine distance, and a past-only smoothed lane-rate encoding. The 12,000 validation predictions and 31 December predictions are positive and in the required formats.

December curve: 821.17 − 842.36 (mean 832.02); *largestadjacent* − *daychange*10.47. The lane, distance (360 mi), equipment (Dry Van), and weight (32,000 lb) are fixed; only date changes.



Validation performed

score.py accepted all 12,000 validation rows and 31 December rows and generated the chart above. This script validates structure, IDs, dates, and positive values; it does not calculate the Spotter leaderboard metrics. Those scores must be obtained after submission.

Limitations: coordinate consistency is internal to the supplied data and does not verify city locations against an external source. The large road-mile versus haversine discrepancy is expected to include road routing and should not be ?corrected? without a trusted distance source. R-squared is not included in the saved assessment metrics.